



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	RAZIA BANU
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Roll Number	2520030534
Branch	CSE
Course Outcome	CO2

Case Study Topic: HDFC NetBanking Daily Spend Analysis using Fenwick Tree (Set 3 Q2)

Work Done Summary:

In this case study, a Fenwick Tree (Binary Indexed Tree - BIT) was implemented to efficiently analyze and manage daily spending records. The primary objective was to perform fast calculations of cumulative expenditures and spending within specific date ranges without repeatedly traversing the entire dataset. Daily transaction amounts were stored in the BIT structure, where each update operation efficiently modified the relevant nodes. Using the Fenwick Tree, prefix sums representing total spending up to a particular date were calculated, and range-sum queries were performed by combining prefix sum results.

To ensure accuracy, the outputs generated by the Fenwick Tree were compared with manually calculated totals, and the results matched successfully. The implementation demonstrated how the Fenwick Tree significantly improves performance by reducing both update and query operations from $O(n)$ in a traditional array-based approach to $O(\log n)$. This efficiency makes Fenwick Trees highly suitable for real-time financial systems, banking applications, expense tracking platforms, and data analytics environments where frequent updates and rapid retrieval of cumulative financial information are required. The case study highlights the practical advantages of using advanced data structures to handle large volumes of transactional data efficiently and accurately.

Implementation Details:

1. Constructed Binary Indexed Tree.
2. Performed point updates.
3. Calculated prefix sums.
4. Executed range query from day 5 to day 12.
5. Verified answer ₹7650.

Result: Screenshots / Output

```
PS D:\Coding\Projects\Case Studies> cd "d:\Coding\Projects\Case Studies" ; javac FenwickSpendAnalysis.java ; java FenwickSpendAnalysis
HDFC NETBANKING DAILY SPEND ANALYSIS

Daily Spending Records:
1200 800 0 2400 1500 600 0 0 3500 0 1100 950 700 0

Prefix Sum(12) = 12050
Prefix Sum(4) = 4400

Query:
Total Spending from Day 5 to Day 12

Result:
Range Sum(5,12) = ₹7650

OBSERVATIONS
- Fenwick Tree stores cumulative sums efficiently.
- Prefix sums are computed in  $O(\log n)$ .
- Range queries are answered using two prefix sums.
- Suitable for banking and financial analytics systems.

TIME COMPLEXITY
Update Operation :  $O(\log n)$ 
Prefix Sum Query :  $O(\log n)$ 
Range Sum Query :  $O(\log n)$ 
Space Complexity :  $O(n)$ 
PS D:\Coding\Projects\Case Studies>
```

Prefix Sum(12) = 12050

Prefix Sum(4) = 4400

Range Sum(5,12) = ₹7650

Update Operation : $O(\log n)$

Prefix Sum Query : $O(\log n)$

Range Sum Query : $O(\log n)$

Space Complexity : $O(n)$

GITHUB LINK : <https://github.com/P-Shashe-Preetham/DSA-CASE-STUDIES.git>

Student Signature	Faculty Signature